# AHMAD ZAKI JULIAN

*Computer Engineering Graduate • IoT & Embedded Systems • Network Infrastructure*

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## PROFESSIONAL SUMMARY

Diploma III graduate in Computer Engineering from Sriwijaya University with a GPA of 3.78/4.00 and a strong interest in Information Technology, networking, troubleshooting, and IoT development. Familiar with microcontroller programming, computer maintenance, network installation, and system integration through academic and internship experience. A fast learner, adaptable, and collaborative individual with strong problem-solving skills and a motivation to continuously improve technical abilities.

## PROFESSIONAL EXPERIENCE

**Sriwijaya University** — Agust 2023 – September 2023
*Internship — Computer Laboratory* — *Palembang, South Sumatra*

- Conducted basic troubleshooting on laboratory equipment.
- Assisted in IoT-based Smart Hydroponic project development, including hardware assembly, sensor testing, and microcontroller integration.

**PT Bukit Asam Tbk** — January 2024 – February 2024
*Internship — Mine Control Center* — *Tanjung Enim, South Sumatra*

- Configured and distributed HT radio devices for mining operations personnel.
- Monitored radio communication systems from the Mine Control Center to ensure operational continuity.
- Performed signal analysis and supported repeater optimization using Motorola SLR1000 in mining tunnel areas.

**PT Taruna Anugrah Mandiri (TAM)** — July 2025 – October 2025
*Field Network Technician* — *Palembang, South Sumatra*

- Installed end-to-end LAN network infrastructure across multiple client sites, including structured cable pulling, RJ45 termination and crimping, patch panel setup, and connectivity testing — ensuring network reliability from day one of handover.
- Deployed Fiber Optic (FO) cable installations with proper routing, splicing preparation, and termination to deliver high-speed, low-loss connectivity in line with client specifications.
- Executed full-cycle CCTV installation projects: camera mounting and alignment, cable routing and concealment, DVR/NVR configuration, and remote access setup — delivering functional security surveillance systems for clients.
- Coordinated with project supervisors and clients on-site to ensure installations met technical standards, timelines, and safety requirements.

## EDUCATION

**Sriwijaya University** — 2020 – 2024
*Associate Degree (A.Md.Kom.), Computer Engineering* — *Palembang, South Sumatra*

- GPA: 3.78 / 4.00
- Focused on IoT development, microcontroller programming, networking, and system integration.

**SMA Bukit Asam** — 2016 – 2019
*Science Track (IPA)* — *South Sumatra*

## RESEARCH & ACADEMIC PROJECT

**Undergraduate Thesis — Sriwijaya University**
*Design and Implementation of Automatic Irrigation System Based on Fuzzy Logic & Internet of Things in Smart Hydroponic Showcase* *Passed — 2024*

- Designed and developed an automatic hydroponic irrigation system using Mamdani Fuzzy Logic and Internet of Things (IoT) technology to improve irrigation efficiency and automate water circulation control.
- Integrated NodeMCU ESP8266 microcontroller with HC-SR04 water level sensor, YF-S401 water flow sensor, relay modules, and Blynk IoT platform for real-time monitoring and remote system control.
- Developed a fuzzy logic algorithm capable of automatically adjusting mixer speed based on water level and water flow rate input data to maintain optimal hydroponic conditions.
- * Conducted hardware integration, sensor calibration, troubleshooting, and real-time communication testing between sensors, microcontroller, and IoT dashboard systems.
- *Achieved responsive irrigation control performance during testing with water level inputs ranging from 0–6 cm and water flow rates between 7.1 ml/s and 9.5 ml/s.
- Recorded mixer speed output of 162 RPM at 2 cm water level and 7.1 ml/s flow rate with an average error rate of 0%, while at 4 cm water level and 9.5 ml/s flow rate the system produced 74 RPM with an average error rate of 0.5%.
- Successfully maintained stable hydroponic irrigation conditions with accurate sensor readings and low system error, demonstrating high effectiveness and reliability for smart agriculture applications.
- Prepared complete technical documentation including system architecture design, fuzzy logic implementation, sensor performance analysis, testing methodology, and overall system evaluation.

## TECHNICAL SKILLS

**Electrical & Infrastructure:** LAN structured cabling; RJ45 crimping & termination; Fiber Optic (FO) installation; CCTV system installation & configuration (DVR/NVR); HT radio device configuration; repeater installation

**IoT & Embedded Systems:** NodeMCU ESP8266, Arduino, sensor integration, HC-SR04 ultrasonic sensor, YF-S401 water flow sensor, relay module integration, IoT-based monitoring systems, and Blynk IoT platform.

**Software & Tools:** Microsoft Office — Word, Excel, PowerPoint (technical reporting, data analysis, project presentations)

**Core Competencies:** Field installation & commissioning; Technical troubleshooting; Safety-conscious site work; Technical report writing; Cross-team coordination

**Languages:** Indonesian (Native) • English (Passive — reading & written comprehension)

## ADDITIONAL INFORMATION

**Mobility:** Willing to be placed anywhere in Indonesia; experienced working in field-based and remote environments (mining area, multi-site installation projects).

**Interests:** Telecommunications, electrical power systems, network infrastructure, and embedded systems.

**References:** Available upon request.